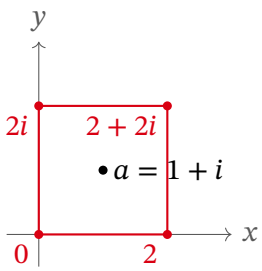
# Rotations and complex numbers

Advanced high school · geometry · Lesson and method · 09

**Prerequisites.** Algebraic form, modulus and coordinates.

**Key point · Ideas and method.** The complex number $z = x + iy$ represents $(x, y)$. Multiplying by $i$ gives $iz = -y + ix$, which represents $(-y, x)$: a counterclockwise quarter-turn about the origin. Multiplication by $-1$ gives a half-turn, and by $-i$ a clockwise quarter-turn. They preserve distances from the origin because $|i| = 1$.
For rotation about centre a, translate to the origin, rotate and translate back: $z' = a + i(z - a)$. This counterclockwise rotation preserves distances between all points.

**Example 1.** With $a = 1 + i$ and $z = 3 + 2i$, $z - a = 2 + i$ and $i(z - a) = -1 + 2i$. The image is $z' = 3i$. Relative vectors $(2, 1)$ and $(-1, 2)$ are perpendicular and both have norm $\sqrt{5}$.




**Activity 1 · Explore.** In the complex-number lab, move z and compare z with iz. Check equal moduli and perpendicular vectors.